

Scalar-angular torsion: A geometric and dimensionally consistent formulation for the unification of spacetime, quantum mechanics, and particle dynamics

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The Scalar–Angular–Torsion (SAT) theory, also known as SAT/Blockwave, presents a unified geometric framework that rigorously integrates the Standard Model, General Relativity, and Quantum Mechanics through a minimal geometric ontology. Reality is modeled as a 4D network of particle worldlines, termed **filaments**, propagating through an advancing **time surface**. Fundamental constants, mass, and force laws emerge solely from the topology and geometry of the filament-timesheet intersection. Adhering to the **Zero-Parameter Economy** (Normalization Principle), the theory axiomatically introduces **zero numerical parameters**; all dimensionless constants are structurally derived from internal geometry. An iterative calibration process successfully converged on a dimensionless coupling ratio $\mathbf{R} \approx 0.007304$, matching the fine-structure constant (α) with high precision. We present the **Total Covariant Action** (S_{SAT}), the rigorously derived Euler–Lagrange system, and the emergent **Topological Mass Suppression Law** ($\mathbf{m} \propto 1/\mathbf{Q}$). Stability constraints, termed the **UV Finiteness Lock**, mandate that only particle configurations with a Topological Charge $\mathbf{Q} \leq 3$ can be dynamically stable ground states in 4D spacetime. Based on this constraint, we predict the existence of two stable, novel ground-state particles: $\mathbf{Q} = 2$ ($\approx 0.5073 \times 10^{-27}$ kg) and $\mathbf{Q} = 3$ ($\approx 0.3403 \times 10^{-27}$ kg), which serve as primary candidates for dark matter.

I. INTRODUCTION AND FOUNDATIONAL ONTOLOGY

The Scalar-Angular Torsion (SAT) theory proposes a minimal geometric ontology consisting of three fundamental primitives: the **Filament** (the 4D particle worldline), the **Time Surface** (Σ_t , the advancing 3D wavefront), and the **Particle** (the localized intersection point where the filament pierces the time surface). This framework aims to translate established physics (Standard Model, General Relativity, and Quantum Mechanics) into equivalent geometric language, achieving synthesis through intersectional geometry.

A. Axiomatic Constraints: The Zero-Parameter Economy

A core commitment of the framework is the **Zero-Parameter Economy** (Normalization Principle). This principle dictates the axiomatic introduction of **zero numerical parameters** into the fundamental structure, requiring all fundamental dimensionless constants and physical ratios to be derived solely from the system’s internal geometry and topology. This constraint is utilized as the ultimate test of the framework’s honesty. The curvature observed in General Relativity (GR) is geometrically attributed to the collective “back-pull” exerted by the filament structure on the time surface.

II. FORMAL MECHANICS AND DYNAMICS

A. Total Covariant Action

The dynamics of the system are determined by the **Total Covariant Action** (S_{SAT}), which incorporates both continuous fields (ξ) and discrete topological configurations (τ) via a generalized partition function (Z_{SAT}). The action is structured to import standard physics Lagrangians ($\mathcal{L}_u, \mathcal{L}_{\text{gauge}}, \mathcal{L}_{\text{Dirac}}$) alongside a **Topological Action** (S_{Top}).

The framework centrally features the **discrete twist charge** (τ), formally defined as a structure ‘TopoInv’ containing the winding number (L_{wind}), the linking number (L_{link}), and the writhe number (W_{writhe}), all natural numbers ($\mathbb{N}$). The total **Topological Charge** Q is the sum of these invariants:

$$Q(\tau) = \tau \cdot L_{\text{wind}} + \tau \cdot L_{\text{link}} + \tau \cdot W_{\text{writhe}} \quad (1)$$

Topological coupling is explicitly incorporated in action definitions, such as the **BF Topological Action** (S_{BF}), which depends on $Q(\tau)$.

B. Euler–Lagrange System and Formal Verification

The continuous fields include the scalar (θ), vector (u), emergent spinor (ψ), and filament current (J). The non-linear nature of the theory is apparent in the specified Euler–Lagrange equations, such as the one governing the scalar field θ :

$$\nabla(\nabla\theta^\mu)\mu x + 3\sin(3\theta^\mu x) = 0 \quad (2)$$

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The term $3\sin(3\theta^\mu x)$ uses standard LaTeX math operators; based on the source notation, the term ‘Real.sin’ is assumed to be the symbolic function sin. The non-relativistic hydrodynamic limit detailing conservation laws is also provided.

Formal verification is undertaken using the `proof_constraints` theorem within the Lean 4 computational environment to ensure internal mathematical stability across the mass spectrum. This process validates necessary constraints including **Positive Effective Mass** ($m_{\text{eff}} > 0$), **Topological Dominance** ($R < 1$), and **Positive Topological Charge** ($Q > 0$). The formalization in Lean 4 suggests that physics may be reconstructible within a proof-assistant logic, gesturing toward **semantic unity**.

III. EMERGENT MECHANISMS AND NOVEL PREDICTIONS

A. Topological Mass Mechanism

The SAT framework replaces the conventional Higgs mechanism by positing mass as an emergent property of geometric tension and topology. The total observable mass ($\mathbf{m}_{\text{eff}}$) is defined as the sum of the dominant **Topological Mass** (m_{topo}) and the secondary **Induced Inertial Mass** (m_{ind}).

The central tenet of the mass mechanism is the **Topological Mass Suppression Law**, which states that topological mass is inversely proportional to the Topological Charge $Q(\tau)$:

$$\mathbf{m}_{\text{topo}} = \mathbf{m}_0 / Q \quad (3)$$

This law implies that **greater topological complexity yields smaller mass**.

The Induced Mass (m_{ind}) arises from dynamic drag against bosonic excitations called **Temporons** (δu^μ), resulting from fluctuations of the unit time-flow vector (u^μ). Particle stability relies on the **Topological Dominance Condition** ($R \equiv m_{\text{ind}}/m_{\text{topo}} \ll 1$).

B. Geometric Gauge Structure and Calibration

Particle identity is determined geometrically: **Fermions** are identified as structural filaments carrying helices, yielding half-integer spin; **Bosons** are defined as dynamic torsional patterns or parasitic ripples propagating along the fermionic substrate. The

Strong Interaction is geometrically explained by a $\mathbb{Z}_3$ fusion rule enforced by topological twist fields, modeling QCD’s three-quark structure.

The iterative calibration process successfully converged on key empirical anchors:

1. The Hadronic Mass Scale ($Q = 1$) was determined: $\mathbf{m}_{\text{eff}} \approx \mathbf{1.0073} \times 10^{-27}$ kg.
2. The dimensionless coupling ratio converged to $\mathbf{R} \approx \mathbf{0.007304}$, which matches the empirical fine-structure constant (α).

Consistency checks confirmed the calculated W-to-Z boson mass ratio and ensured the derived numerical gravitational constant (G_{ind}) matches the known empirical value:

C. Falsifiability and Dark Matter Predictions

The theory imposes a critical stability requirement, the **UV Finiteness Lock**, which mandates that only particle configurations with a Topological Charge $\mathbf{Q} \leq \mathbf{3}$ can exist as dynamically stable ground states in 4D space-time.

The mass formula predicts two novel stable ground-state particles ($N = 0$) based on this constraint, proposed as dark matter candidates:

- $\mathbf{Q} = \mathbf{2}$ state: $\approx \mathbf{0.5073} \times 10^{-27}$ kg
- $\mathbf{Q} = \mathbf{3}$ state: $\approx \mathbf{0.3403} \times 10^{-27}$ kg

The detection of these specific, stable, low-mass particles constitutes the primary test of the framework. Unstable excited states ($N \geq 1$) are predicted to follow a quadratic mass-squared spectrum formula: $\mathbf{m}^2 = \mathbf{k}(\mathbf{N} + \beta \mathbf{Q}^2 - \mathbf{a})$.

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The theory’s claim to **aesthetic closure** relies fundamentally on the **Zero-Parameter Economy** as evidence of mathematical inevitability.

IV. SUPPLEMENTAL MATERIAL

In line with Physical Review D submission guidelines, all extensive derivations, proofs, and calibration data are relegated to Supplemental Material. This material includes the Lean 4 code necessary for verification of the core constraints.